MAHATMA GANDHI UNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG4DSCECT204 - Computer Organization

Duration:1.5 Hrs. Maximum Marks:50

Part A

Very Short Answer Questions
Answer **All** Questions. Each question carries **1** mark

1. Write the binary equivalent of the decimal number 68.	U CO1
2. Name the memory that has the fastest access time.	K CO1
3. Identify the addressing mode that requires two memory accesses.	K CO1
4. State the component used to interface memory with the CPU.	U CO1
5. Find the 1's complement of the binary number 100110.	A CO2
6. Identify a valid BCD number.	K CO2
7. Write the result of the Boolean expression A + A′.	U CO2
8. State the source from which control signals are generated in a microprogrammed control unit.	K CO2
9. State the output condition of an RS flip-flop when R = 1 and S = 0.	U CO3
10. Name the register used in stack addressing mode.	K CO3

1 x 10 = 10

Part B

Short Answer Questions
Answer **4** Questions. Each question carries **5** marks

11. Compare XOR and XNOR gates.?	K CO2
12. Explain the concept of Memory Hierarchy with a neat diagram.	U CO1
13. Simplify the Boolean expression using Karnaugh Map:F(A,B,C,D)=Σm(0,2,5,7,13,15).	A CO3
14. Explain the working principle of a Master-Slave JK flip-flop with a neat diagram.	U CO3
15. Describe Cache Memory.	U CO1
16. Illustrate the difference between Hardwired control and Microprogrammed control.	A CO2

4 x 5 = 20

Part C

Essay Questions
Answer any **2** Questions. Each question carries **10 m**arks

17. Describe the mapping techniques used in cache memory.	K CO1
18. Construct the logic diagrams and truth tables for RS, JK, T, and D flip-flops, and explain their operation with the help of sequential circuit concepts.	A CO3
19. Explain the hardwired control and microprogrammed control of a processor unit.	U CO2

2 x 10 = 20